

RISHIK MUTHYALA

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Education

University of Massachusetts Amherst

Sep. 2023 – May 2026

Bachelor of Science in Computer Science & Mathematics, Minor in Business — GPA: 3.9/4.0

Amherst, Massachusetts

Relevant Coursework

- Data Structures
- Algorithms
- Object Oriented
- Discrete Mathematics
- Artificial Intelligence
- Probability and Statistics
- Calc I, II
- Linear Algebra
- Multivariate Calculus
- Differential Equations
- Accounting
- C Programming

Experience

SelfServe

January 2025 – Present

iOS Developer

Boston, MA

- Architecting and developing the company's flagship iOS application from the ground up, using SwiftUI for UI/UX implementation and core functionality development, which reduced time-to-market by 40%
- Designing the entire backend infrastructure and API architecture, using Amazon Web Services and PostgreSQL, to enable real-time analytics and guest management capabilities for hospitality clients across 100+ properties

Treevah

December 2024 – Present

Software Development Engineer Intern

Chicago, Illinois

- Engineered comprehensive file management system using React.js, implementing hierarchical subfolder organization and advanced sorting algorithms, resulting in 40% improvement in file retrieval efficiency
- Developed sophisticated search functionality with dynamic filtering and contextual right-click operations, enhancing user interaction and reducing average file access time by 25%
- Architected backend infrastructure using Azure services, implementing robust user authentication, RESTful API endpoints, and scalable data retrieval systems
- Led financial modeling initiatives for feature development costs and cloud infrastructure, creating comprehensive ROI analyses and optimization strategies
- Authored detailed Confidential Information Memorandum (CIM) documentation, translating technical features into business value propositions for stakeholders and potential investors

Drobot Co.

May 2023 – September 2023

Software Engineer Intern

Boston, MA

- Developed autonomous drone prototype using Python, TensorFlow, and PyTorch, improving object tracking accuracy by 30% and implemented real-time MAVLink communication system for seamless data transfer between control station and drone
- Optimized mapping algorithms, reducing processing time by 40% and designed custom object detection model accuracy for multi-object tracking
- Utilized CAD for drone component prototyping, reducing weight by 25% and increasing flight time by 15%
- Created data analytics pipeline with Pandas and NumPy, processing 1TB+ of flight data; integrated sensor fusion algorithms, enhancing drone stability and precision by 35%

Projects

FoundU (2024 Hackathon Winner) | React Native, Node.js, Express, Google Vision API, Puppeteer **November 2024**

- Engineered lost UCard recovery system serving 30,000+ UMass students, enabling finders to photograph cards, with Google Vision API extracting student information and custom Puppeteer automation scraping UMass finder directory
- Developed streamlined user flow using React Native for cross-platform interface, Node.js/Express backend for secure data processing, and Nodemailer for automated location-based email notifications

Leadership

First Tech Robotics

June 2022 – June 2023

Co-Captain

Woonsocket, RI

- Co-captained and programmed Java-based competitive robot, implementing sensor fusion and PID algorithms to optimize autonomous task execution in dynamic environments

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, Swift, HTML, CSS

Frameworks/Libraries: React, React Native, Node, Express, Next, SwiftUI, TensorFlow, PyTorch, Redux, Pandas, NumPy, WebSockets, OAuth, Azure SDK, Puppeteer, Google Vision API

Tools: VS Code, Xcode, Android Studio, Git, GitHub, Docker, Arduino, Linux, Figma

Cloud/Database: Amazon Web Services, Google Cloud Platform, Azure, MongoDB, SQL, PostgreSQL